

1. In this problem, we consider sending real-time voice from Host A to Host B over a packet-switched network (VoIP). Host A converts analog voice to a digital 128 kbps bit stream on the fly. Host A then groups the bits into 64-byte packets. There is one link between Host A and B; its transmission rate is 4 Mbps and its propagation delay is 8 msec. As soon as Host A gathers a packet, it sends it to Host B. As soon as Host B receives an entire packet, it converts the packet's bits to an analog signal. How much time elapses from the time a bit is created (from the original analog signal at Host A) until the bit is decoded (as part of the analog signal at Host B)? (note: $128 \text{ kbps} = 128 \times 10^3 \text{ bps}$, $4 \text{ Mbps} = 4 \times 10^6 \text{ bps}$)

A) 2.08 msec
B) 6.08 msec
C) 9.128 msec
D) 12.128 msec

2. a) Consider a packet of length L which begins at end system A and travels over three links to a destination end system. These three links are connected by two packet switches. Let d_i , s_i , and R_i denote the length, propagation speed, and the transmission rate of link i , for $i = 1, 2, 3$. The packet switch delays each packet by d_{proc} . Assuming no queuing delays, in terms of d_i , s_i , and R_i ($i = 1, 2, 3$), and L , what is the total end-to-end delay for the packet?

A) $L/(R_1 + R_2 + R_3) + d_1/s_1 + d_2/s_2 + d_3/s_3 + 2d_{\text{proc}}$
B) $L/R_1 + L/R_2 + L/R_3 + d_1/s_1 + d_2/s_2 + d_3/s_3 + 2d_{\text{proc}}$
C) $LR_1 + LR_2 + LR_3 + d_1s_1 + d_2s_2 + d_3s_3 + 2d_{\text{proc}}$
D) $LR_1 + LR_2 + LR_3 + d_1s_1 + d_2s_2 + d_3s_3$

b) Suppose now packet is 1,500 bytes, the propagation speed on all three links is $2.5 \times 10^8 \text{ m/s}$, the transmission rates of all three links are 2 Mbps, the packet switch processing delay is 3 msec, the length of the first link is 5,000 km, the length of the second link is 4,000 km, and the length of the last link is 1,000 km. For these values, what is the end-to-end delay? (Note that $M = 10^6$, $k = 10^3$)

A) 24 msec
B) 64 msec
C) 86 msec
D) 96 msec

5. The `tracert` command (in Windows) or `traceroute` command (in Mac/Unix/Linux) is one of the key diagnostic tools for TCP/IP. It displays a list of all the routers that a packet must go through to get from the computer where `tracert` is run to any other computer on the Internet. To use the command, type the `tracert` (or `traceroute`) command followed by the host name of the computer to which you want to trace the route in the command prompt. E.g., `tracert cityu.edu.hk` OR `traceroute cityu.edu.hk`. Answer the following questions:
- What is the meaning of each column in the traceroute output?
 - Why do we see " *" sometimes?
 - Why is the round-trip delay of router N sometimes longer than the round-trip delay of router N+1 (e.g., the delays to Router 3 are larger than the delays to Router 4)?

3. In the above problem, suppose $R_1 = R_2 = R_3 = R$ and $d_{\text{proc}} = 0$. Further suppose the packet switch does not store-and-forward packets but instead immediately transmits each bit it receives before waiting for the entire packet to arrive. What is the end-to-end delay?

A) 16 msec
B) 26 msec
C) 36 msec
D) 46 msec

4. A packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, one other packet is halfway done being transmitted on this outbound link and four other packets are waiting to be transmitted. Packets are transmitted in order of arrival. Suppose all packets are 1,500 bytes and the link rate is 2 Mbps. What is the queuing delay for the packet? (Note that $M = 10^6$.)

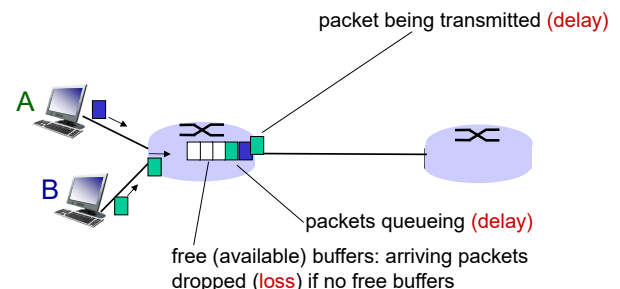
A) 27 msec
B) 37 msec
C) 47 msec
D) 57 msec

More generally, what is the queuing delay when all packets have length L , the transmission rate is R , x bits of the currently-being-transmitted packet have been transmitted, and n packets are already in the queue?

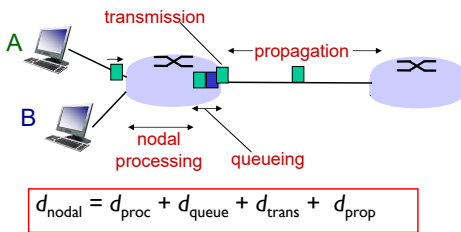
How do loss and delay occur?

packets *queue* in router buffers

- ❖ packet arrival rate to link (temporarily) exceeds output link capacity
- ❖ packets queue, wait for turn



Four sources of packet delay



d_{proc} : nodal processing

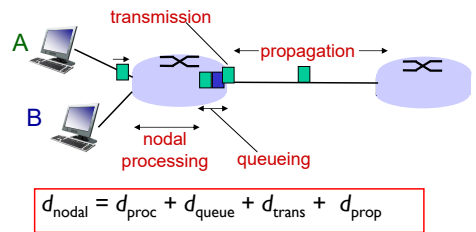
- check bit errors
- determine output link
- typically < msec

d_{queue} : queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

Introduction 1-2

Four sources of packet delay



d_{trans} : transmission delay:

- L : packet length (bits)
- R : link bandwidth (bps)
- $d_{\text{trans}} = L/R$

d_{prop} : propagation delay:

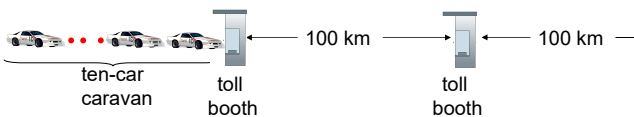
- d : length of physical link
- s : propagation speed in medium ($\sim 2 \times 10^8$ m/sec)
- $d_{\text{prop}} = d/s$

d_{trans} and d_{prop} very different

* Check out the Java applet for an interactive animation on trans vs. prop delay

Introduction 1-3

Caravan analogy



- cars "propagate" at 100 km/hr
- toll booth takes 12 sec to service car (bit transmission time)
- car ~ bit; caravan ~ packet
- Q: How long until caravan is lined up before 2nd toll booth?

- time to "push" entire caravan through toll booth onto highway = $12 \times 10 = 120$ sec
- time for last car to propagate from 1st to 2nd toll booth: $100 \text{ km} / (100 \text{ km/hr}) = 1$ hr
- A: 62 minutes

Introduction 1-4

Q1

- Consider the first bit, before it can be transmitted, all of the bits in the same packet must be generated. This requires

$$\frac{64 \times 8}{128 \times 10^3} = 4 \text{ msec}$$

- The time required to transmit the packet is

$$\frac{64 \times 8}{4 \times 10^6} = 128 \text{ microsec}$$

- Propagation delay is 8 msec.
- The delay until decoding is 4 msec + 128 microsec + 8 msec = 12.128 msec.
- A similar analysis shows that all bits experience a delay of 12.128 msec.

Q2

- The first end system requires L/R_1 to transmit the packet onto the first link;
- the packet propagates over the first link in d_1/s_1 ; the packet switch adds a processing delay of d_{proc} ;
- after receiving the entire packet, the packet switch connecting the first and the second link requires L/R_2 to transmit the packet onto the second link;
- the packet propagates over the second link in d_2/s_2 . Similarly, we can find the delay caused by the second switch and the third link: L/R_3 , d_{proc} , and d_3/s_3 .

Q2

- Adding these eight delays gives

$$d_{end-end} = L/R_1 + L/R_2 + L/R_3 + d_1/s_1 + d_2/s_2 + d_3/s_3 + d_{proc} + d_{proc}$$

- To answer the second question, we simply plug the values into the equation to get $6 + 6 + 6 + 20 + 16 + 4 + 3 + 3 = 64$ msec.

Q3

- Because bits are immediately transmitted, the packet switch does not introduce any delay; in particular, it does not introduce a transmission delay. Thus,

$$d_{end-end} = L/R + d_1/s_1 + d_2/s_2 + d_3/s_3$$

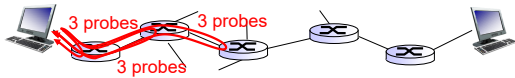
- For the values given, we get $6 + 20 + 16 + 4 = 46$ msec.

Q4

- The arriving packet must first wait for the link to transmit $4.5 * 1,500$ bytes = 6,750 bytes or 54,000 bits.
- Since these bits are transmitted at 2 Mbps, the queuing delay is 27 msec.
- Generally, the queuing delay is $(nL + (L - x))/R$.

“Real” Internet delays and routes

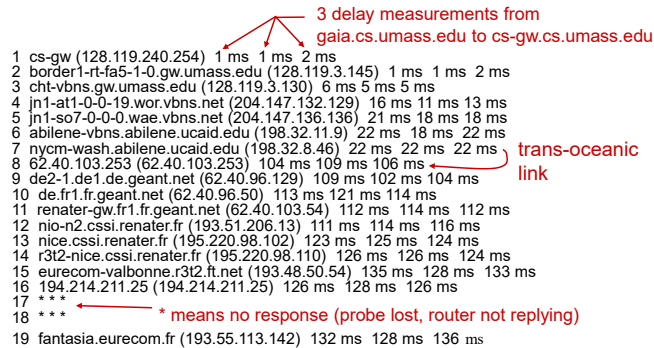
- ❖ what do “real” Internet delay & loss look like?
- ❖ **traceroute** program: provides **delay measurement** from source to router along end-end Internet path towards destination. For all i :
 - sends three packets that will reach router i on path towards destination
 - router i will return packets to sender
 - sender times interval between transmission and reply.



Introduction 1-10

“Real” Internet delays, routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr



Introduction 1-11

Q5

- IP address of the routers between the source and the destination, three round-trip time measurements for each router
- An asterisk is displayed if a packet is not acknowledged within the expected timeout (The default timeout is 5 sec). Timeout appears mainly because a firewall blocks the traffic.

Q5

- The round-trip delays include all of the delays, including transmission delays, propagation delays, router processing delays, and queuing delays. Because the queuing delay is varying with time, the round-trip delay of packet N sent to a router N can sometimes be longer than the round-trip delay of packet $N+1$ sent to router $N+1$.